ILLINOIS TECH | College of Computing

Ch10: The Digital Divide, Democracy, And Work P4:Cybertechnology & Workplace

93

Announcements

- Internet 02 section- clarification
- Final Exam

94

Learning Objectives

Upon completion of this lesson the students should be able to:

- Define the Digital Divide
  - Describe and discuss the moral and ethical issues that define and surround the Digital Divide and proposals for its resolution
- Describe and discuss issues surrounding making technology available to people with disabilities
- Discuss the effect of technology on racism
- Discuss the impact of technology on gender issues
  - Describe and discuss gender issues in employment in the technology field
- Explain and discuss the impact of technology on democracy and democratic values
- Describe how technology has transformed the nature of work and discuss its impact on the quality of work life

95

Module Objectives

Upon completion of this lesson the students should be able to:

- Describe how technology has transformed the nature of work and discuss its impact on the quality of work life

96

3. Social Sectors: The Workplace

- We have already examined some impacts that cybertechnology has had for two (of three) broad social categories:
- *sociodemographic groups* (affecting social/economic class, race, gender, and disabled populations),
- *social and political institutions* (affecting democracy).
- We next examine some impacts that cybertechnology has had for a third (and final social category) – *social sectors*.
- In particular, we focus on some impacts affecting one critical sector: employment and work.

97

Employment and Work

- We examine two different kinds of concerns affecting the impact of cybertechnology on employment and work:

1. the *quantity* of jobs resulting from the use of cybertechnology;
2. the *quality* of work-life for employees in the digital era.

98

Cybertechnology, the Quantity of Jobs and the Transformation of Work

- Thus far, it is not clear whether cybertechnology has either created or eliminated more jobs.
- It is clear, however, that cybertechnology has *transformed* (the nature of) work in the digital era.

99

Cybertechnology and the Trans-formation of Work

- Those arguing that cybertechnology has reduced jobs point to the number of factory and assembly jobs that have been automated (and thus eliminated).
- Those claiming that cybertechnology has created jobs, argue that it has introduced newer industries, such as computer support companies.
- This *transformation*, or shift in jobs overall, can be analyzed in terms of two broad categories:

I. *job displacement* and automation,
II. *globalization* and *outsourcing*.

100

Job Displacement and Automation

- *Job displacement* can be measured in terms of the net result of jobs gained and lost.
- Job displacement involving workers has been significantly affected by *automation*, where human workers were replaced by machines (beginning with the Industrial Revolution in the 19th century).
- "Luddites" (followers of Ned Ludd) reacted to auto- mation in the textile industry by smashing machinery.
- People who oppose new technology in the workplace today are sometimes referred to as "NeoLuddites."

101

Robotics (and Some Implications for Job Displacement)

- Developments in the field of *robotics* have also raised social concerns affecting job displacement.
- *Robots* have already replaced or displaced many factory and "blue collar" workers.
- Consider that some robots have been programmed to perform tasks that are:

a) routine and mundane for humans,
b) considered hazardous to humans.

102

Robotics

- Patrick Lin (2012) notes that robots (in work-related contexts) are typically tasked to perform the "three Ds" — i.e., jobs that humans consider "dull, dirty, and dangerous."
- Although robots were once fairly unsophisticated, contemporary robotic systems are now able to perform a wide range of tasks in the workplace.
- Some ethical aspects of robots and robotic systems are examined in detail in Chapter 12.

103

Expert Systems (and Some Impli-cations for Job Displacement)

- While (physical) robots have eliminated many blue-collar jobs, sophisticated programs called *expert systems* increasingly threaten many professional jobs.
- An expert system (ES) is a problem-solving computer program that is "expert" at performing one particular task.
- ESs use "inference engines" to capture the decision-making strategies of experts (usually professionals).
- They execute instructions that correspond to a set of rules an expert would use in performing a professional task.
- "Knowledge engineer" ask human experts in a given field a series of questions to extract the critical rules (used by professionals in their field of work) and then they design programs based on the responses to those questions.

104

Expert Systems

- Initially, expert systems were designed to perform jobs in chemical engineering and geology, both of which required the professional expertise of highly educated persons and were also generally considered too hazardous for humans.
- More recently, however, ESs have been developed for use in professional fields such as law, education, and finance.
- ES programs have replaced some highly-skilled professional workers in a number of areas or fields.
- Examine the end-of-chapter scenario by Forester and Morrison (in the textbook), which raises some ethical questions about designing an "expert administrator" who may need to "lie" or to mislead people in order to be an *expert at* its task.

105

Remote Work

- One way that cybertechnology has transformed work for many employees is by making it possible for them to work "remotely" — i.e., outside the traditional workplace.
- Even though remote work is a relatively recent practice, it has already raised some social and ethical questions.
- One question has to do with whether all employees who perform remote work benefit from it equally.
- For example, are white-collar employees affected in the same way as employees who are less-educated and less-skilled?
- It is one thing to be a white-collar professional with an option to work at home at your discretion and convenience, but it is very different for some clerical, or "pink collar," workers who may be required to work remotely out of their homes.

106

Remote Work

- Some professional men and women may *choose* to work at home because of child-care considerations or because they wish to avoid a long and tedious daily commute.
- However, other employees may not have the same options, especially those in lower-skilled jobs, because their employers may require them to work at home.
- In some case, people required to work remotely may not have the same opportunities for promotions and advance- ments as their (more visible) counterparts who have the option of working in a traditional workplace setting.
- So, employees in some situations may be disadvantaged because of employer-specific, remote-work policies.

107

Job Outsourcing

- Initially, the *outsourcing* of jobs mainly affected employees in manufacturing-related industries.
- Outsourcing now affects many highly-skilled "white collar" jobs as well, including some high-tech jobs.
- For example, some programming jobs traditionally held by employees in American companies are now outsourced to companies in India and China.
- Ironically, the jobs of programmers, whose skills were essential to make (the current practice of) remote work a reality, are now being outsourced to countries where programmers earn less money.

108

Globalization

- In addition to, and often in connection with, outsourcing and the loss of jobs, the economies of some nations have also been severely impacted by *globalization*.
- Monahan (2005) defines globalization as

  ...the blurring of boundaries previously held as stable and fixed...between local/global, public/private [and] nation/world.

109

Globalization

- Discussions of globalization tend to focus on concerns affecting:
- labor outsourcing,
- international trade agreements,
- immigration,
- cultural homogenization.

110

Globalization

- Our concern is mainly with the economic aspects of globalization, particularly as they impact cybertechnology and the workplace.
- Consider that trade agreements have made possible a new global economy, encouraging:
- greater competition between nations,
- greater efficiency for businesses.
- It is unclear whether these trade agreements have economically benefited some countries.

111

The *Quality* of Work-life

- Quality issues include concerns about employee health, which can pertain both to physical-and-mental-health related issues.
- Two quality-related concerns involving the contemporary workplace result from:

1. *employee stress* and *workplace surveillance*;
2. *computerized monitoring* (including the monitoring of an employee's electronic devices).

112

Employee Stress and Workplace Surveillance

- A 2008 report on employee monitoring by the American Management Association noted that
- 43% of American companies monitor employee email,
- 96% of those companies "track external (incoming and outgoing) messages."
- The report also noted that 45% of companies track the amount of time an employee spends at the keyboard.

113

Employee Stress and Workplace Surveillance

- An increasing number of these companies now also monitor the blogosphere to see what is being written about them in various blogs, as well as what is said about them on social networking sites such as Facebook.
- As a result of increased monitoring, many employees have been fired for misusing a company's email resources or its Internet connection, or both.
- Some practices involving workplace monitoring have also contributed to increased employee stress.

114

The Expansion of Workplace Surveillance

- Kizza and Ssanyu (2005) describe some factors that have contributed to the recent expansion and growth of employee monitoring, two of which are the:

1) plummeting prices of both software and hardware,
2) miniaturization of monitoring products.

- The lower cost of software and hardware has made monitoring tools available to many employers who, in the past, might not have been able to afford them.
- The miniaturization of monitoring tools has made it much easier to conceal them from employees.

115

The Expansion of Workplace Surveillance

- Introna (2004) points out that surveillance technology, in addition to becoming less expensive, has also become "less overt and more diffused."
- He also notes that current monitoring technologies have created the potential to build surveillance features into the "very fabric of organizational processes."

116

The Expansion of Workplace Surveillance

- To support Introna's view, consider that monitoring tools are now being used to measure such things as the:
- number of minutes an employee spends on the telephone completing a transaction (such as selling a product);
- number of minutes an employee spends on the telephone completing a transaction (such as booking a reservation);
- number and length of breaks an employee takes.

117

Workplace Surveillance and Monitoring of Electronic Devices

- Examine Scenario 10-3 (the *Ontario v Quon* case) in the textbook, which describes some controversies involving the monitoring of an employee's electronic pager.
- Was Jeff Quon's privacy violated in this case?
- Did Quon have a reasonable expectation of privacy in this particular incident, as the lower court initially ruled?
- Should there be any limitations or constraints placed on an employer's right to monitor an employee's conversations on electronic devices?
- Or, should all forms of employee monitoring be permis- sible, where employer-owned equipment is involved?

118

Computerized Monitoring: Some Important Distinctions

- Weckert (2005) points out that it is crucial to draw some distinctions involving two areas of computerized monitoring, which affect:

1) the different *applications of monitoring*,
2) the different *kinds of work situations*.

119

Computerized Monitoring: Some Distinctions

- Regarding the *different kinds of applications*, Weckert notes that employees could be monitored with respect to the following kinds of activities:
- email usage,
- URLs visited while Web surfing,
- quality of their work,
- speed of their work,
- work practices (health and safety),
- employee interaction.

120

Computerized Monitoring: Some Distinctions

- With regard to the *different kinds of work situations*, Weckert believes that some further distinctions also need to be made.
- He notes that it may be appropriate to monitor the keystrokes of data-entry workers to measure their performance in specific period of time.
- Weckert also notes that it may not be appropriate to mon- itor email in cases where client confidentially is expected.
- For example,, a therapist employed in a health organization may receive highly sensitive and personal e-mail from one of her client's regarding the client's mental state or physical health.

121

Rationales Used to Support Computerized Monitoring

- Those who support computerized monitoring in the workplace tend to believe that it:
- improves workplace productivity;
- improves corporate profits;
- guards against industrial espionage;
- reduces employee theft.

122

Rationales Used to Oppose Computerized Monitoring

- Those who oppose computerized monitoring in the workplace tend to believe that it:
- increases employee stress;
- invades employee privacy;
- reduces employee autonomy;
- undermines employee trust.

123

Table Summarizing Some Rationales for Supporting and for Opposing Monitoring

Rationales Used Support to Monitoring	Rationales Used to Oppose Monitoring
Improves worker productivity	**Increases employee stress**
Improves corporate profits	**Invades employee privacy**
Guards against industrial espionage	**Reduces employee autonomy**
Reduces employee theft	**Undermines employee trust**

124

International Dimensions of Compu-terized Monitoring in the Workplace

- Do we now need international agreements for employee-monitoring policies (involving cybertechnology) because of the global workforce?
- Coleman (2005) notes that an employees' privacy could be violated by software-monitoring programs that reside on a computer located in a country different from the one in which the person is (physically) working.
- Coleman also suggests that an International Bill of Human Rights should be adopted to address global aspects of the employee monitoring.

125

?

126